

Shourov Joarder

✉ joardershourov60@gmail.com

🌐 shourovj

🔄 shourovj

🌐 shourovj

📀 shourovj

Education

Bangladesh University of Engineering & Technology (BUET)

Bangladesh

B.Sc. in Electrical and Electronic Engineering

Feb 2020 – March 2025

Major: Communication and Signal Processing (CSP)

CGPA: 3.88/4.00 (2nd in CSP major)

Relevant Courses: Artificial Intelligence and Machine Learning | Digital Image Processing I | Computer Programming | Linear Algebra | Probability and Statistics | Random Signals and Processes | Microprocessors and Embedded System | Digital Signal Processing | Control Systems | Digital Electronics

Research Interests

Multimodal LLM | VLMs | Reinforcement Learning | Autonomous Vehicle

Publications

1. **Shourov Joarder**, D. Sikdar, A. Akash, B. Battarai, P.K. Gyawali, “Two is better than one: A Collapse-free Multi-Reward RLIF Training Framework,” Submitted to Neurips’26 ([Preprint](#)).
TLDR: This work focuses on using **intrinsic reward** signals to guide reasoning paths to mitigate mode collapse and improve self-correction of LLMs for better decision-making agents with limited supervision.
2. **Shourov Joarder**, T. Talukder, K. Hasan, “Multi-Stage Residual-Aware Unsupervised Deep Learning Framework for Consistent Ultrasound Strain Elastography,” Ready for Submission ([Preprint](#)).
TLDR: Developed a multi-stage **unsupervised spatio-temporal** training framework for deep learning based ultrasound strain imaging. This work has been validated for open source simulation, **in vivo** and our own **clinical** datasets.
3. S. Hasan, **Shourov Joarder**, A. Nayem, H. Hasan, and S. A. Fattah, “Multilingual Voice-Controlled Smart Wheelchair with Advanced Features,” Published at [IEEE ECCE](#). (DOI: [10.1109/ECCE64574.2025.11013785](#))
TLDR: Developed a full scale wheelchair integrated with multilingual voice-control, collision detection and avoidance.
4. A. Dhar, D. Sikder, A. Shovon, and **Shourov Joarder**, “Skin Cancer Semantic Segmentation,” Published at [IEEE ECCE](#). (DOI: [10.1109/ECCE64574.2025.11013785](#)).
TLDR: Unet-based Stacked Hourglass model converts cartesian image to polar image which is then fed to a **TransUnet** model for estimating the semantic segmentation on skin cancer **ISIC** dataset.

Research Experiences

Research Collaborator, West Virginia University (WVU)

Mar 2026 - May 2026

Supervisor : [Prof. Prashnna K. Gyawali](#)

Multi-reward Unsupervised RLIF for LLM Reasoning

Designed a multi-reward RL framework using intrinsic reward signals to guide diverse reasoning paths in LLMs, mitigating mode collapse and improving self-correction with limited supervision.

- Proposed complementary intrinsic rewards to prevent reward hacking during on-policy training; validated on multiple reasoning benchmarks.
- Paper submitted to NeurIPS’26.

Undergraduate Thesis Student, EEE, BUET

Mar 2024 - Mar 2025

Supervisor : [Dr. Kamrul Hasan](#)

Multi-Stage Unsupervised Sequential Deep Learning Method for Ultrasound Strain Elastography

Developed MUSSE-Net, an unsupervised sequential network for displacement and strain estimation, validated on simulation, **in vivo**, and clinical datasets.

- Designed novel *Contextual-Aware Feature Encoder*, *TriCrossAttention*, and *Cross-Attentive Sequential Decoder* modules; implemented in PyTorch.
- Outperforms state-of-the-art baselines [ReUSENet](#), [USENet](#) on SNR, CNR, and NRMSE metrics.

Ongoing Researches

• **Risk-aware Multi-agent Reinforcement Learning**

May 2026 - Present

This work focuses on the collaboration of multi-agent systems. How agents collaborate with each other to finish a work effectively.

• Self-Distillation of Robust VLM

May 2026 - Present

Working on a cross-input self-distillation framework in which a single VLM serves as both teacher and student and stabilises on-policy training.

• Autonomous Driving with Efficient VLM-based reasoning with LoRA Distillation ([Github](#))

The goal of this research is to develop a robust and explainable system for Autonomous Driving (AD) for real time application. I have implemented a Trajectory and path planning for AD with explainability and reasoning. Initially, used **nexar-dashcam AD dataset** and opensource VLM model (Qwen).

Work Experience

Adjunct Lecturer, Dept. of CSE, [BRAC University](#).

June 2025 - Present

Courses taken: *Image Processing (CSE428), Digital Electronics (CSE350)*

Machine Learning Engineer (Part-time), [ACI Ltd.](#)

Apr 2025 - Present

Ongoing Projects

- **Medical-ExpertVLM** with explainability and personalized assistance for doctors and medical students.
- Developing a robust OCR and annotation tool using state-of-the art **Qwen2.5VL-VLM** to extract medicine details and bank cheque details from images of a handwritten prescription and cheques.
- Developing an **AI-Assisted** System for **Drug Discovery** and Development

Honors and Awards

- University Merit Scholarship (3 times) 2020, 2021, 2023
- University Dean's List Scholarship (2 times) 2021, 2022
- University Stipend (2 times) 2021, 2022
- 1st Runner-Up of Undergraduate Project Idea Contest at 25th ICCIT 2022, Bangladesh. [[Certificate](#)]
- 57th in the public leaderboard of [DL Sprint](#) - BUET CSE Fest 2024, Bengali AI Math Olympiad a LLM based competition.
- **Best Notebook Award** at the [DL Sprint](#) - BUET CSE Fest 2022, Bengali ASR Competition.

Selected Projects

Coding Google's PaliGemma VLM from Scratch [Github](#)

Implemented Google's opensource **PaliGemma** Vision Language Model (VLM) from scratch. Implemented the **SigLip** vision encoder, **KV-cache** and the **GemmaLM** with the multimodal projector. This project was inspired by Umair Jamil's paligemma open-source project.

Autonomous Inventory Robot [Github](#)

Developed an autonomous robotic system capable of executing real-time voice commands to identify, retrieve, and transport specified objects. The system integrates **Google Speech API** based speech recognition, **YOLOv5-s** for real-time object detection with robotic arm manipulation for precise object grasping with a **Raspberry-pi** as the processor. Following successful acquisition, the object is returned to a predefined base location using a Line Following Robot (**LFR**) navigation system.

Deep-Learning-based-Breast-Cancer-Classification-Using-VGGIN [Github](#)

Trained the **VGGIN** model—a custom deep learning architecture that integrates **VGG-19** with the **Inception** module, on the BreakHis histopathology dataset. Achieved a test accuracy of 99.628%, demonstrating the model's effectiveness in classifying breast cancer subtypes from histopathological images.

Voice Controlled Wheelchair for Disabled Patients [Github](#)

Developed a full scale voice-controlled wheelchair for physically impaired people. **GMM**-based trained **VoiceRecognitionV3** module takes voice commands from the patients in any language and in any accent and moves accordingly. In addition, the wheelchair collision avoidance and emergency help feature.[[Video](#)].

Extracting Audio from Muted Video [Github](#)

The main goal of this project was to extract the audio signal from a muted video using signal processing methods in MATLAB. The local and global pixel motions were captured using **Complex Steerable Pyramid** decomposition. This was originally a project by Abe Davis, MIT [[Visual Microphone](#)].

Skills

ML, DL Framework	PyTorch, TensorFlow, Transformers, vLLM, LangChain
Programming	Python, MATLAB, C/C++, Linux, Verilog, Assembly, LaTeX, Git
Simulations and Tools	Raspberry Pi, Arduino, Proteus, Pspice, Quartus, Keil
Frontend Backend	FastAPI, Flask, Gradio, Android Studio, Kotlin (Basic).